

Magnets

MCQ

1. A magnet was first discovered by a shepherd named

- a) Aristotle
- b) Magnus
- c) Galileo
- d) Newton

Answer: b) Magnus

2. The natural magnet discovered by Magnus was found in

- a) A river
- b) A forest
- c) A mountain
- d) A desert

Answer: c) A mountain

3. The rock that attracted iron materials is known as

- a) Limestone
- b) Sandstone
- c) Magnetite
- d) Granite

Answer: c) Magnetite

4. Magnetite is also called

- a) Marble stone
- b) Magnetic stone
- c) Precious stone
- d) Heavy stone

Answer: b) Magnetic stone

5. Natural magnets are also known as

- a) Artificial stones
- b) Lodestones
- c) Iron ores
- d) Steel rocks

Answer: b) Lodestones

6. The place believed by some to be the discovery site of magnets is
- a) Athens
 - b) Rome
 - c) Magnesia
 - d) Sparta

Answer: c) Magnesia

7. Which of the following is a property of natural magnets?
- a) They have fixed shapes
 - b) They are temporary
 - c) They occur naturally
 - d) They repel iron

Answer: c) They occur naturally

8. Natural magnets are permanent magnets because
- a) They lose magnetism easily
 - b) They occur naturally
 - c) They do not need electricity
 - d) They do not lose magnetism easily

Answer: d) They do not lose magnetism easily

9. The process of converting iron into a magnet is called
- a) Heating
 - b) Charging
 - c) Magnetization
 - d) Levitation

Answer: c) Magnetization

10. Man-made magnets are called
- a) Natural magnets
 - b) Artificial magnets
 - c) Permanent rocks
 - d) Iron stones

Answer: b) Artificial magnets

11. Which of the following is an artificial magnet?

- a) Lodestone
- b) Magnetite
- c) Bar magnet
- d) Magnetic rock

Answer: c) Bar magnet

12. A horseshoe magnet is also called

- a) Ring magnet
- b) U-shaped magnet
- c) Disc magnet
- d) Needle magnet

Answer: b) U-shaped magnet

13. Which material is attracted by a magnet?

- a) Plastic
- b) Paper
- c) Iron
- d) Rubber

Answer: c) Iron

14. Which of the following is a magnetic material?

- a) Gold
- b) Silver
- c) Nickel
- d) Leather

Answer: c) Nickel

15. Which metal is non-magnetic?

- a) Iron
- b) Nickel
- c) Cobalt
- d) Gold

Answer: d) Gold

16. Iron filings are

- a) Dust particles

- b) Sand particles
- c) Small pieces of iron
- d) Soil particles

Answer: c) Small pieces of iron

17. Iron filings are mainly used to

- a) Make magnets
- b) Show magnetic field
- c) Remove rust
- d) Clean magnets

Answer: b) Show magnetic field

18. The property by which magnets attract materials is called

- a) Reflection
- b) Refraction
- c) Attraction
- d) Magnetism

Answer: d) Magnetism

19. The needle in the experiment is attracted because

- a) It is light
- b) It is sharp
- c) It is magnetic
- d) It is made of iron

Answer: d) It is made of iron

20. The parts of a magnet where attraction is maximum are called

- a) Centre
- b) Middle
- c) Poles
- d) Sides

Answer: c) Poles

21. A magnet has how many poles?

- a) One
- b) Two

c) Three

d) Four

Answer: b) Two

22. The pole that points towards the north is called

a) South pole

b) East pole

c) West pole

d) North pole

Answer: d) North pole

23. Magnetic force is maximum at the

a) Centre

b) Sides

c) Poles

d) Surface

Answer: c) Poles

24. Magnetic force is minimum at the

a) Poles

b) Centre

c) Ends

d) Edges

Answer: b) Centre

25. When a magnet is freely suspended, it rests in

a) East–West direction

b) North–South direction

c) South–East direction

d) Random direction

Answer: b) North–South direction

26. The direction-finding property of magnets is used in

a) Cooking

b) Painting

c) Navigation

d) Farming

Answer: c) Navigation

27. The Chinese used lodestones mainly for

a) Decoration

b) Navigation

c) Medicine

d) Toys

Answer: b) Navigation

28. A device used to find directions is called

a) Bar magnet

b) Electromagnet

c) Compass

d) Generator

Answer: c) Compass

29. The red end of the compass needle shows

a) South direction

b) East direction

c) West direction

d) North direction

Answer: d) North direction

30. Mountaineers use a compass to

a) Measure height

b) Find direction

c) Predict weather

d) Find food

Answer: b) Find direction

31. Which material is used to make a magnet at home?

a) Plastic nail

b) Wooden stick

c) Iron nail

d) Rubber strip

Answer: c) Iron nail

32. While making a magnet, the bar magnet should be moved

a) Back and forth

b) In circular motion

c) In one direction

d) Randomly

Answer: c) In one direction

33. Repeating the rubbing process helps in

a) Heating the nail

b) Breaking the nail

c) Magnetizing the nail

d) Bending the nail

Answer: c) Magnetizing the nail

34. A homemade compass works because of

a) Gravity

b) Air pressure

c) Magnetism

d) Heat

Answer: c) Magnetism

35. In a homemade compass, the needle floats on

a) Oil

b) Water

c) Sand

d) Air

Answer: b) Water

36. Like poles of magnets

a) Attract

b) Repel

c) Rotate

d) Break

Answer: b) Repel

37. Unlike poles of magnets

- a) Repel
- b) Ignore
- c) Attract
- d) Melt

Answer: c) Attract

38. Repulsion occurs when

- a) N–S poles face
- b) S–N poles face
- c) N–N poles face
- d) Magnet is broken

Answer: c) N–N poles face

39. Attraction occurs when

- a) Like poles face
- b) Unlike poles face
- c) Magnets are heated
- d) Magnets are dropped

Answer: b) Unlike poles face

40. Which action can destroy magnetism?

- a) Cooling
- b) Polishing
- c) Heating
- d) Painting

Answer: c) Heating

41. Magnetism can be lost by

- a) Washing
- b) Hammering
- c) Painting
- d) Drying

Answer: b) Hammering

42. Dropping a magnet from a height causes

- a) More magnetism
- b) No change
- c) Loss of magnetism
- d) Colour change

Answer: c) Loss of magnetism

43. Bar magnets should be stored

- a) Singly
- b) In water
- c) In pairs
- d) Near TV

Answer: c) In pairs

44. While storing bar magnets, unlike poles should be

- a) Opposite
- b) On the same side
- c) Separated
- d) Removed

Answer: b) On the same side

45. Soft iron pieces used for storage are called

- a) Covers
- b) Keepers
- c) Blocks
- d) Shields

Answer: b) Keepers

46. Magnets should be kept away from

- a) Books
- b) Clothes
- c) Electronic devices
- d) Plants

Answer: c) Electronic devices

47. A crane in a junkyard uses

- a) Natural magnet

- b) Temporary magnet
- c) Electromagnet
- d) Plastic magnet

Answer: c) Electromagnet

48. Magnets are used in loudspeakers to

- a) Produce light
- b) Produce sound
- c) Store data
- d) Increase weight

Answer: b) Produce sound

49. Which device uses magnets for navigation?

- a) Calculator
- b) Compass
- c) Thermometer
- d) Barometer

Answer: b) Compass

50. Electric motors work based on

- a) Gravity
- b) Friction
- c) Magnetism
- d) Heat

Answer: c) Magnetism

51. Levitation means

- a) Falling down
- b) Floating without support
- c) Rotating fast
- d) Sliding

Answer: b) Floating without support

52. Magnetic levitation is due to

- a) Attraction
- b) Repulsion

c) Gravity

d) Heat

Answer: b) Repulsion

53. When like poles are placed together, magnets

a) Stick

b) Break

c) Levitate

d) Melt

Answer: c) Levitate

54. Electromagnetism occurs when

a) Magnet is heated

b) Iron is cooled

c) Current flows through a coil

d) Magnet is dropped

Answer: c) Current flows through a coil

55. An electromagnet works only when

a) It is cooled

b) Current flows

c) It is permanent

d) It is painted

Answer: b) Current flows

56. Changing the direction of current changes

a) Shape

b) Colour

c) Poles

d) Size

Answer: c) Poles

57. A train working on magnetic levitation is called

a) Diesel train

b) Bullet train

c) Maglev train

d) Metro train

Answer: c) Maglev train

58. Maglev trains move mainly due to

a) Gravity

b) Friction

c) Repulsion

d) Heat

Answer: c) Repulsion

59. One advantage of Maglev trains is

a) More noise

b) More pollution

c) High speed

d) High friction

Answer: c) High speed

60. Maglev trains are eco-friendly because

a) They use coal

b) They use petrol

c) They produce no pollution

d) They move slowly

Answer: c) They produce no pollution

61. Maximum speed of Maglev trains can reach up to

a) 100 km/h

b) 200 km/h

c) 300 km/h

d) 600 km/h

Answer: d) 600 km/h

62. Which country uses Shanghai Maglev train?

a) Japan

b) India

c) China

d) Korea

Answer: c) China

63. SC Maglev train is used in

a) China

b) Japan

c) India

d) Korea

Answer: b) Japan

64. KTX Maglev train is used in

a) South Korea

b) China

c) Japan

d) India

Answer: a) South Korea

65. Which of the following routes is proposed for Maglev in India?

a) Delhi–Kolkata

b) Mumbai–Delhi

c) Chennai–Trichy

d) Kochi–Trivandrum

Answer: b) Mumbai–Delhi

66. Magnetite contains mainly

a) Plastic

b) Rubber

c) Iron

d) Gold

Answer: c) Iron

67. Which object uses magnets in daily life?

a) Pencil

b) Notebook

c) Refrigerator sticker

d) Eraser

Answer: c) Refrigerator sticker

68. Door locks using magnets are

- a) Manual
- b) Mechanical
- c) Magnetic
- d) Wooden

Answer: c) Magnetic

69. Phone covers use magnets for

- a) Decoration
- b) Locking
- c) Colouring
- d) Heating

Answer: b) Locking

70. Which is NOT a use of magnets?

- a) Navigation
- b) Lifting loads
- c) Cooking food
- d) Making toys

Answer: c) Cooking food

71. Which shape magnet has two ends close together?

- a) Bar magnet
- b) Ring magnet
- c) Horseshoe magnet
- d) Disc magnet

Answer: c) Horseshoe magnet

72. A magnetic needle is free to rotate inside

- a) Clock
- b) Compass
- c) Motor
- d) Generator

Answer: b) Compass

73. Which of the following is non-magnetic?

- a) Cobalt
- b) Iron
- c) Nickel
- d) Rubber

Answer: d) Rubber

74. Magnetic field can be shown using

- a) Sand
- b) Iron filings
- c) Water
- d) Plastic

Answer: b) Iron filings

75. Magnets are NOT used in

- a) Loudspeakers
- b) Motors
- c) Compasses
- d) Thermometers

Answer: d) Thermometers

76. A magnet attracts materials made of

- a) Wood
- b) Iron
- c) Glass
- d) Plastic

Answer: b) Iron

77. Which magnet has a hole in the centre?

- a) Bar magnet
- b) Ring magnet
- c) Horseshoe magnet
- d) Needle magnet

Answer: b) Ring magnet

78. A magnet loses its strength if stored

- a) Properly

- b) With keepers
- c) Near electronics
- d) In pairs

Answer: c) Near electronics

79. The study of magnets is called

- a) Biology
- b) Chemistry
- c) Magnetism
- d) Astronomy

Answer: c) Magnetism

80. Which property makes magnets useful for centuries?

- a) Colour
- b) Weight
- c) Direction-finding
- d) Shape

Answer: c) Direction-finding